

MINI ASSIGNMENT

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Topic: API Error Handling and Safe JSON Access

1. If an API returns 429, what should your program do?

A **429 Too Many Requests** response indicates that the program has sent requests faster than the API allows. The program should temporarily stop making requests and try again after some time. It can use a **delay or exponential backoff** before retrying. If the API provides a **Retry-After** value, the program should follow it. A proper message should also be shown to the user.

2. If `response.json()` fails, what should your program do?

The program should not assume that every API response is valid JSON. It should handle the JSON parsing error using proper **error handling** instead of allowing the application to crash. The program can first check the response status and content type. If the response contains invalid JSON, HTML, or plain text, it should handle the situation safely and display an appropriate error message.

3. If a JSON key is missing, why might `.get()` be safer than `[]`?

Using `[]` to access a key that does not exist can cause a **KeyError**, which may stop the program. The `.get()` method is safer because it returns **None or a specified default value** when the key is unavailable. For example, `data.get("email", "Not Available")` returns "Not Available" if the email key is missing. Therefore, `.get()` helps make API programs more reliable.

Conclusion: Good API handling requires the program to manage rate limits, invalid responses, and missing data properly. Using retries, exception handling, and safe JSON access prevents unexpected crashes and makes the application more reliable.